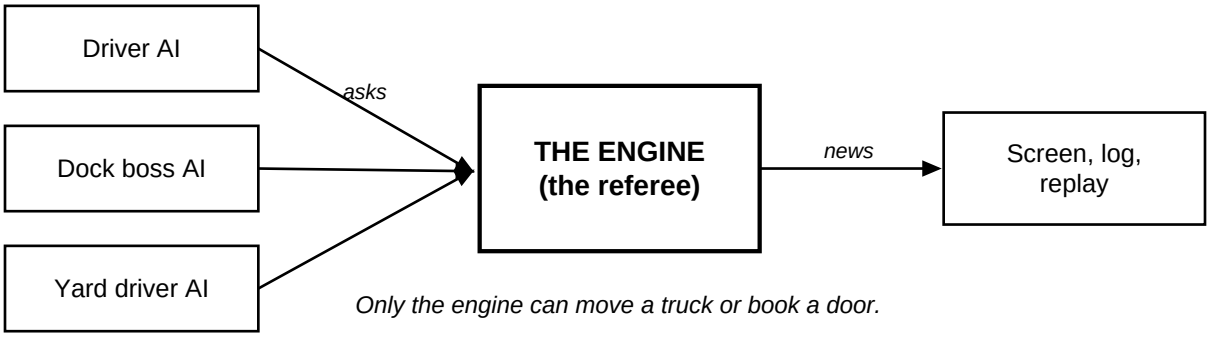
The Engine, Explained Simply

Warehouse Sim · design draft · October 2026. Black-and-white guide to the part that keeps the simulation honest.

1. What the engine is

Think of a board game. The players (our AI truck drivers, dock bosses and yard drivers) talk, bargain and make plans. The **engine is the referee and the game board**. It is the only thing allowed to move a truck, book a door or count pallets. Players can only *ask* it to do things. It checks the rules and says yes or no.



Only the engine can move a truck or book a door.

Three promises the engine always keeps:

It never guesses. If we don't know a number (say, how many forklifts the real building has), it is stored as *unknown*, never as 0 or a made-up value. Any action that needs that number is refused until someone supplies it.

It never half-does anything. A request either fully happens or doesn't happen at all.

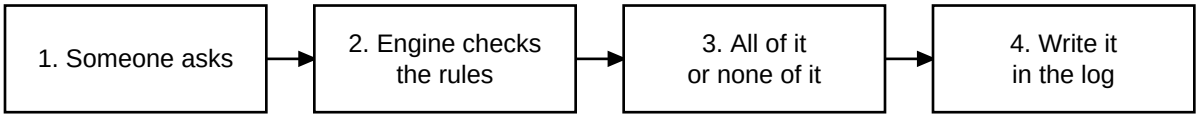
It gives the same answer every time. Same starting point plus same requests gives exactly the same result. That is what makes replay possible.

2. What it keeps track of

Thing	What the engine remembers	Example
The clock	Pretend time, which only moves when the engine says so	Day 1, 14:07:30
The building	Every door, ramp, parking stall and staging lane (counts from the published specs)	182 doors, 4 ramps, 235 stalls
Equipment and crews	Yard tractors, forklifts and crews, with counts from *your* scenario file	6 yard tractors (your input)
Trucks and trailers	Where each one is, and whether a trailer is attached to a truck	Trailer T-2291 at door D-108
Visits	Each truck's trip: arriving, waiting, at a door, leaving	Visit V-047, inbound
Cargo	Pallets, and where they are right now	22 pallets on T-2291
Deals	Who agreed to what, for which time window	D-112 from 14:20 to 15:05
Version number	Goes up by one every time anything changes	v3815

3. How anything changes: ask, check, all-or-nothing, record

Nobody edits the world directly. Every change is a **command** that goes through the same four steps:



Example: a driver needs a new door

1. Driver DRV-047 asks: "Book door D-112 for me from 14:20 to 15:05."

2. The engine checks: Is D-112 open? Is it free then? Is a crew and a forklift free then? Is the driver's info up to date?

3. Everything checks out, so it books the door, the crew and the forklift **together**. If any one of them were busy, it would book **none** of them.

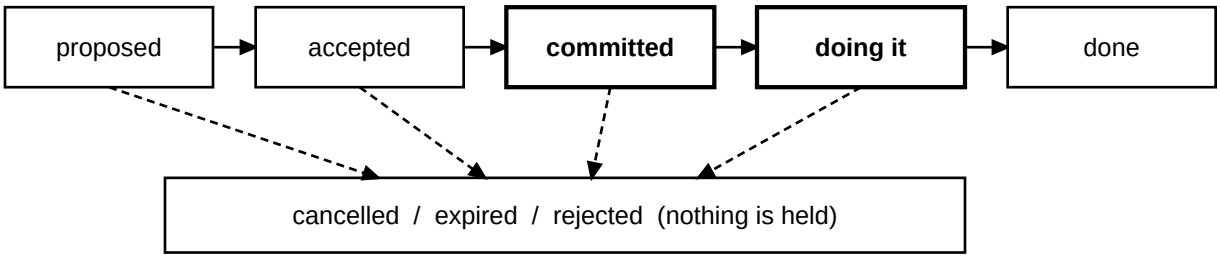
4. It writes "deal C-88 committed, version 3815" into the log. The screen and replay read from that log.

Every command carries a **request ID**. If the same request arrives twice (for example, the network retried it), the engine recognises the ID and returns the first answer. Nothing gets booked twice.

4. Deals: how agents agree

Agents don't book things alone. They make **deals**, which the engine calls commitments. A deal moves through these stages:

Only the two bold boxes hold a door, crew or forklift.



A deal only **holds** resources once it is **committed**. Talking, offering and even agreeing reserve nothing. A promise in a chat message is not a booking.

> **Example: two drivers want the same door**
>
> Driver A and Driver B both try to commit door D-112 for 14:20–15:05 at the same moment.
>
> The engine sorts them in a fixed, fair order (time, then priority, then request ID). Driver A comes first and gets the door.
>
> Driver B gets a clear "no": **RESOURCE_CONFLICT**. Driver B can then ask for a different door or time.
>
> Two trucks can never end up booked into one door.

Offers **expire**. If an offer sits unanswered past its deadline, it disappears. If agents keep going round in circles over the same door without agreeing, the engine raises an **escalation**. The supervisor (or a simple first-come-first-served fallback) then settles it, using the same commands as everyone else.

5. When the engine says no

A "no" always comes with a plain reason, and nothing in the world changes.

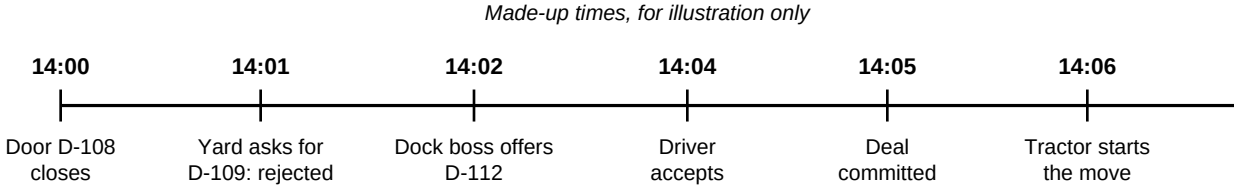
Reason	Plain meaning	Example
STALE_VERSION	You were looking at old information	You asked about D-109, but it was booked a second ago
RESOURCE_CONFLICT	Someone else already has it then	D-112 is booked 14:20–15:05
RESOURCE_UNAVAILABLE	It's closed, broken or absent	Door D-108 is closed
INCOMPATIBLE_DOCK	Wrong kind of door for this trailer	This trailer needs a ramp door
MISSING_CARGO	The pallets aren't where you said	Loading 13 pallets, only 9 staged

Reason	Plain meaning	Example
PRECONDITION_FAILED	Something has to happen first	Can't unload a trailer still at the gate
MISSING_EVIDENCE	We don't know a number this needs	Forklift speed is unknown
NOT_PARTICIPANT / PERMISSION_DENIED	That's not your call to make	A driver tries to close a door
EXPIRED	That offer timed out	The 14:12 deadline passed

6. Time: a pretend clock

The engine keeps its own clock instead of using real time. It keeps a **to-do list of future moments** (a truck arrives, a move finishes, an offer expires, a door reopens) and jumps straight to the next one.

When something important happens, the engine says which agents should wake up and why. **The clock stops while they think.** A slow AI answer never makes a truck late in the simulation. Real thinking time is measured and reported separately.



If two things happen at the same moment, they're handled in a fixed order: disruptions first, then finished work, then expiries, then arrivals, then agent wake-ups.

7. How work actually happens

Agents decide *what* should happen. The engine's built-in rules (the **mechanics**) carry it out at the agreed time.

Task	What it needs	How long it takes
Move a trailer	A free yard tractor and a free destination	12 minutes per move (derived from a USPS audit standard)
Unload or load	A door, a crew and a forklift	pallets ÷ crew speed (crew speed comes from your scenario)
Drop a trailer	A free stall or door	Instant once parked
Hook a trailer	A free tractor at the trailer's spot	Instant once there

Example: unloading time (made-up numbers)

Trailer T-2291 has 22 pallets. Your scenario says a crew moves 30 pallets an hour.

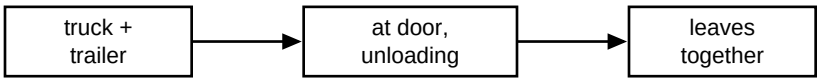
22 ÷ 30 = 0.73 hours, so the unload takes about 44 minutes.

If the door closes after 18 minutes, the engine records 9 pallets done (30 × 0.3 hours, rounded down) and 13 left. When the trailer reaches a new door, it carries on from 9. Nothing is lost and nothing is invented.

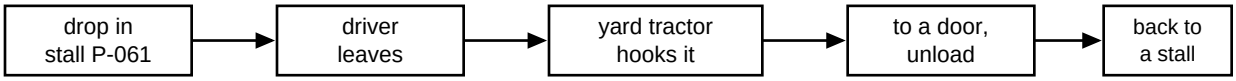
8. Live unload and drop-and-hook

Both happen in real yards, so the engine supports both.

LIVE: the driver stays



DROP-AND-HOOK: the trailer stays, the driver goes



In drop-and-hook the trailer has its own record and stays on site after its driver leaves. That gives yard tractors real work, and it is how a trailer gets moved from door D-108 to D-112 when D-108 closes.

9. Cargo and waiting on other trucks

Pallets are counted, never created or destroyed. Unloaded pallets go to a staging lane. An outbound truck can only leave once all the pallets it needs have been loaded.

Example: one truck waiting on another

Outbound trailer T-2027 at door D-027 needs shipment SH-5512.

SH-5512 is still inside inbound trailer T-2291 (13 pallets left to unload).

So T-2027 can't leave until T-2291 is unloaded and those pallets are loaded onto T-2027. The screen shows this as "blocked by freight".

10. Breaking things on purpose (disruptions)

You can inject problems to see how the agents cope. Each one is labeled **hypothetical** and has a start and an end.

Disruption	What the engine does	Example
Door closes	No new bookings there. Work in progress pauses and keeps its progress. The trailer stays put until it is moved; nothing teleports.	D-108 closed 14:00–16:00
Truck is late	Its arrival time moves. Bookings that depended on it are re-checked and may expire.	V-051 arrives 40 min late
Forklift breaks	Its job is interrupted and the forklift is freed. The job needs another forklift to continue.	F-07 fails at 13:52
Crew is absent	Fewer crews available. Jobs that lose their crew are flagged. Nobody invents a replacement.	Crew CR-3 off from 15:00

11. Rules that can never be broken

After every single change the engine checks these. If one ever fails in testing, it's treated as a bug, never as a result.

Rule	In plain words
One place at a time	A trailer is in exactly one spot. A truck pulls at most one trailer.
One per spot	A door or parking stall holds at most one trailer.
No double-booking	A crew, forklift or door can't be booked for overlapping times.
Broken means unavailable	A closed door or broken forklift can't take new work.
Pallets add up	Pallets on trailers + in staging + shipped always equals the starting total.
All or nothing	A deal books everything it needs, or nothing.
No repeats	The same request ID never takes effect twice.

12. Replay: rewinding the tape

The engine never uses the real clock or random numbers. So if you take the same starting inputs and feed in the same recorded commands, you get **exactly** the same world, down to the last pallet. Replay needs no AI calls at all, so it costs nothing.

> **Example: rehearsing the demo**
>
> Record one full run with the real NVIDIA models on Nebius.
>
> Replay it as many times as you like for practice, for free. The AI is not called again.
>
> To try something different (say, first-come-first-served rules), rewind to a saved moment and branch from there.

13. Not in the first version

These are left out on purpose and will be labeled as simplifications on screen:

- Yard lanes aren't contested, so two moves can't collide.
- Crews are counted as teams, not individual workers.
- Cargo is counted in pallets, without individual product types.

Glossary

Word	Meaning
Agent	An AI player with one job, such as a truck driver or dock boss
Command	A request to the engine. The only way to change anything
Deal / commitment	An agreement between agents that, once committed, books resources
Reservation	A resource booked for a time window, such as door D-112 from 14:20 to 15:05
Mechanics	The engine's built-in rules that carry out agreed work
Escalation	Handing a stuck disagreement to the supervisor
Hypothetical	A problem you injected on purpose; it never happened for real
Unknown	A number we don't have evidence for. It's never filled in with a guess

All times, counts and speeds in the examples are illustrative. Building counts come from the published DFW Commerce Center Building I specifications.